

KARAKALA BHANU PRAKASH REDDY

Software Development Engineer | Bangalore, India

karakalabhanuprakashreddy@gmail.com | +91-7032718222 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Results-driven SDE with 2.5+ years at Oracle building event-driven microservices using Java, Spring Boot, and Kafka. Reduced latency by 550ms, cut CPU by 40%, and processed 5TB/day of JSON workloads in Agile, CI/CD environments.

TECHNICAL SKILLS

Languages: Java, Python, Kotlin, JavaScript, C++, Shell Scripting

Frameworks: Spring Boot, GraphQL, JPA, Hibernate, JBPM, Angular, Maven, Gradle

Tools: Kafka, Docker, JUnit, Karate, Cucumber, Git, CI/CD, AWS, Azure, Linux, Jira

Databases: MySQL, SQL, Redis

Concepts: Microservices, REST APIs, Event-Driven Architecture, OOP, Design Patterns, Agile, Asynchronous Programming

PROFESSIONAL EXPERIENCE

Software Development Engineer

Oct 2023 – Apr 2026

Oracle, Bangalore, India

Credit Card Approval Platform:

- Scaled RESTful microservices backend automating credit-card approvals using Java, Spring Boot, Hibernate/JPA, SQL, and Kafka, reducing decision latency by **550ms**
- Architected REST API layer aligned to SOLID and OOP principles, integrating **5 microservices** with event-driven architecture
- Integrated a **Gen AI**-assisted risk-flagging layer using LLM-based text classification on applicant profiles to surface high-risk cases, reducing manual review effort by **20%**
- Led 5 interns to deliver Smart Comparator PoC using prompt engineering and OpenAI API integration

JSON Relational Duality Views & Vector AI:

- Reduced CPU consumption by **40%** via active consumer – lazy producer model; developed listener component and Broker APIs
- Managed Oracle Data Guard setup reducing recovery time by **30–50%** via optimized switchover and failover
- Benchmarked JSON workloads up to **5TB/day** to evaluate system limits and influence architecture decisions
- Designed and benchmarked an **Embedding-Augmented Generation (EAG)** pipeline integrating **Oracle 23ai Vector Search** with LLM inference APIs, enabling semantic retrieval over large-scale JSON relational data with sub-100ms query latency
- Evaluated **Gen AI** retrieval strategies (RAG vs. EAG, vector similarity vs. keyword) across **50+ sub-projects**, influencing Oracle's internal architectural guidelines for LLM integration with structured databases
- Automated health checks and replication alerts saving the team **3+ hours** of manual work daily

PROJECTS

SmartLend – Loan Approval Microservices System

2025

- Built event-driven loan approval backend using Java, Spring Boot, Kafka, MySQL, and Docker with 3 independent microservices (Loan, Notification, Audit) communicating via Kafka topics
- Designed REST API with JPA persistence, JUnit tests, Swagger UI, and Dockerized deployment | [GitHub](#)

Content Management Web Application

- Automated multi-platform publishing via Instagram and Facebook APIs with secure token-based authentication and Google Drive OAuth2 integration for scheduled media uploads

FidElite – Stock Trading Platform

- Built full-stack trading platform using Angular, Spring Boot, and JPA with real-time portfolio tracking, stock price updates via RESTful APIs, and ACID-compliant Hibernate transaction management

Blockchain-Based Dairy Supply Chain

Apr 2022 – Jun 2022

- Designed Solidity smart contracts to automate validation and payments; conducted 35+ stakeholder interviews translating insights into 12+ MVP features across a decentralized supply chain ledger

LEADERSHIP

Team Lead – Evathon Hackathon | University of Limerick, Ireland

- Ranked **Top 11 out of 350+ global teams**; built facial recognition model using ResNet-50 CNN for exam monitoring and attendance during COVID-19

EDUCATION

Sastra Deemed University – B.Tech Computer Science Engineering

Aug 2019 – Aug 2023

GPA: **8.076/10** | Led peer-learning sessions on OS and Data Structures | Coursework: DSA, OS, DBMS